

SUPPLY CHAIN PERFORMANCE ANALYSIS REPORT

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EXECUTIVE SUMMARY

This report presents a comprehensive analysis of the delivery operations of a global e-commerce company managing end-to-end order fulfillment across multiple regions. The analysis covers 172,765 orders spanning January 2015 through January 2018, with a focus on identifying the root causes of chronic late deliveries, quantifying their financial impact, and establishing a data-driven framework for operational improvement.

The findings reveal that 54.71% of all orders were delivered late, indicating a significant challenge in the company's supply chain and logistics operations. Delayed deliveries resulted in approximately \$2.1 million in profit at risk, highlighting the substantial financial consequences associated with poor delivery performance. Although the company generated \$7.5 million in profit from fulfilled orders, persistent delivery delays continue to erode profitability and negatively impact customer satisfaction.

To address this issue proactively, a Random Forest predictive model was developed to identify orders at risk of late delivery before shipment completion. The model achieved 74% accuracy, demonstrating its potential as an effective decision-support tool for improving delivery performance and reducing operational risk.

- **Key Findings**

Key Finding	Metric
Overall Late Delivery Rate	54.71%
On-Time Delivery Rate	45.29%
Total Orders Analyzed	172,765
Total Profit (Profitable Orders)	\$7.5M
Profit at Risk (Delayed Orders)	\$2.1M
Predictive Model Accuracy (Random Forest)	74%

DATASET OVERVIEW

- **Dataset Source**

The analysis utilizes the DataCo Global Supply Chain Dataset containing e-commerce order transactions, shipment records, customer information, product details, and fulfillment outcomes.

Metric	Value
Records Analyzed	172,765
Time Period	Jan 2015 – Jan 2018
Industry	E-Commerce Retail
Regions Covered	Global
Target Variable	Late delivery risk
Analysis Focus	Delivery Performance & Profitability

BUSINESS CONTEXT & OBJECTIVES

The company operates a global e-commerce platform selling products across categories including sporting goods, fitness equipment, outdoor gear, footwear, apparel, electronics, and consumer products. Orders are fulfilled and shipped across multiple international regions, requiring efficient coordination between inventory management, order processing, and logistics operations.

- **Core Business Problem**

Actual shipping times frequently deviate from scheduled delivery commitments, resulting in :-

- 1) Increased late deliveries and operational inefficiencies.
- 2) Reduced customer satisfaction and trust.
- 3) Lower profitability due to delayed order fulfillment.
- 4) Difficulty in providing reliable delivery commitments.
- 5) Higher risk of customer churn and reputational damage.

As the business scales globally, understanding and mitigating the drivers of delivery delays has become a strategic necessity.

- **Analytical Objectives**

The analysis was designed to:

- 1) Assess delivery performance across regions, departments, customer segments, and shipping modes.
- 2) Quantify the financial impact of delayed deliveries.
- 3) Identify operational bottlenecks causing delays.
- 4) Determine the root causes of delayed shipments.
- 5) Develop a predictive model for identifying high-risk orders.
- 6) Generate actionable recommendations to improve delivery performance and profitability.

KEY PERFORMANCE INDICATORS (KPIs)

The following KPIs were computed to establish a baseline view of delivery performance.

KPI	Value
Total Orders	172,765
Late Deliveries	94,523
Late Delivery Rate	54.71%
On-Time Delivery Rate	45.29%
Total Profit (Profitable Orders)	\$7.5M
Profit at Risk (Delayed Orders)	\$2.1M
90th Percentile Delay	3 Days
Average Order Profit	\$22.03

- **Key Interpretation**

- 1) Late Delivery Rate: 54.71%

More than half of all fulfilled orders were delivered later than scheduled. This indicates that delivery delays are widespread rather than isolated incidents.

- 2) Profit at Risk: \$2.1 Million

Delayed orders collectively place a significant portion of company profits at risk through operational inefficiencies, customer dissatisfaction, and lost future business opportunities.

- 3) 90th Percentile Delay: 3 Days

Ninety percent of delayed orders were late by three days or less, suggesting that delays are generally process-driven rather than caused by catastrophic disruptions.

- 4) Average Order Profit: \$22.03

Each order generates approximately \$22 profit on average, making delivery efficiency critical for maintaining overall profitability.

- **Key Business Insights**

- 1) Late deliveries represent the largest operational challenge identified.

- 2) More than one in two customers experience delayed delivery.

- 3) Delivery delays have substantial financial consequences.

- 4) Delays are typically short in duration but extremely frequent.

- 5) Improving delivery reliability offers significant opportunities for profit protection and customer satisfaction enhancement.

PROFITABILITY ANALYSIS

The profitability analysis evaluates the relationship between delivery performance and financial outcomes.

- **Profitability Distribution**

Order profitability was categorized into:

- 1) Profitable Orders

- 2) Loss-Making Orders

- 3) Break-Even Orders

- 4) Findings

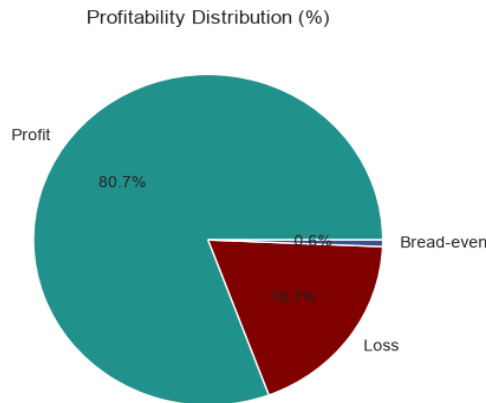
- 5) 80.7% of orders were profitable.

- 6) 18.7% of orders generated losses.
- 7) The remaining orders were break-even.

While the overall business remains profitable, loss-making transactions are disproportionately concentrated among delayed shipments.

Key Insight

Operational inefficiencies causing delivery delays are directly associated with higher rates of financial loss.



- **Delay Distribution & Profit vs. Delay Days**

- 1) Delay Distribution

The analysis reveals:

- a) 31.0% of all orders arrive exactly one day late.
- b) Orders delayed by 1-4 days account for 54.71% of total order volume.
- c) Delays occur consistently across the supply chain rather than through isolated extreme events.

- 2) Profitability by Delay Duration

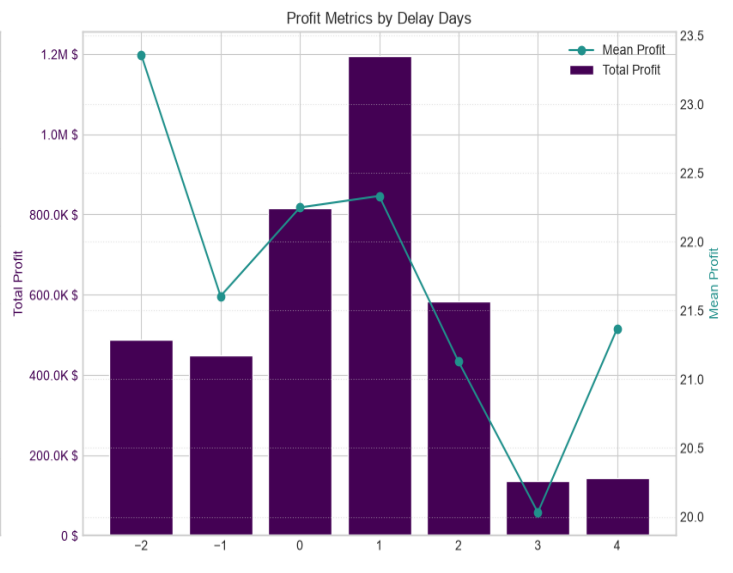
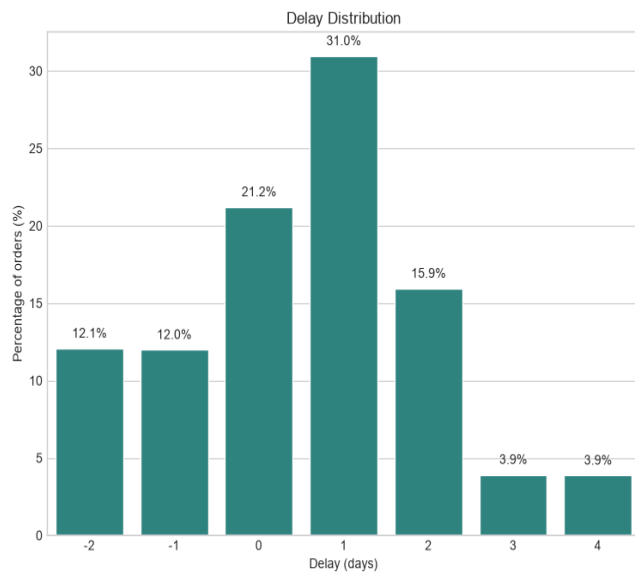
Profit metrics were analyzed across varying delay durations.

Key Finding

Average profit per order remains relatively stable between \$20-\$23 across different delay levels. This indicates that profitability erosion is driven primarily by the volume of delayed orders rather than exceptionally costly individual delays.

Key Insight

Reducing the number of delayed shipments presents a significantly greater financial opportunity than focusing solely on severe delay events.



BOTTLENECK DETECTION

Bottleneck analysis was conducted to determine which operational areas contribute most significantly to late deliveries.

- **Methodology**

Delay percentages were calculated across six operational dimensions:

- 1) Order Region
- 2) Customer Segment
- 3) Shipping Mode
- 4) Order Status
- 5) Order Type
- 6) Department Name

- Delay rate was calculated as:

$$\text{Delay Rate} = \text{Late Orders} \div \text{Total Orders} \times 100$$

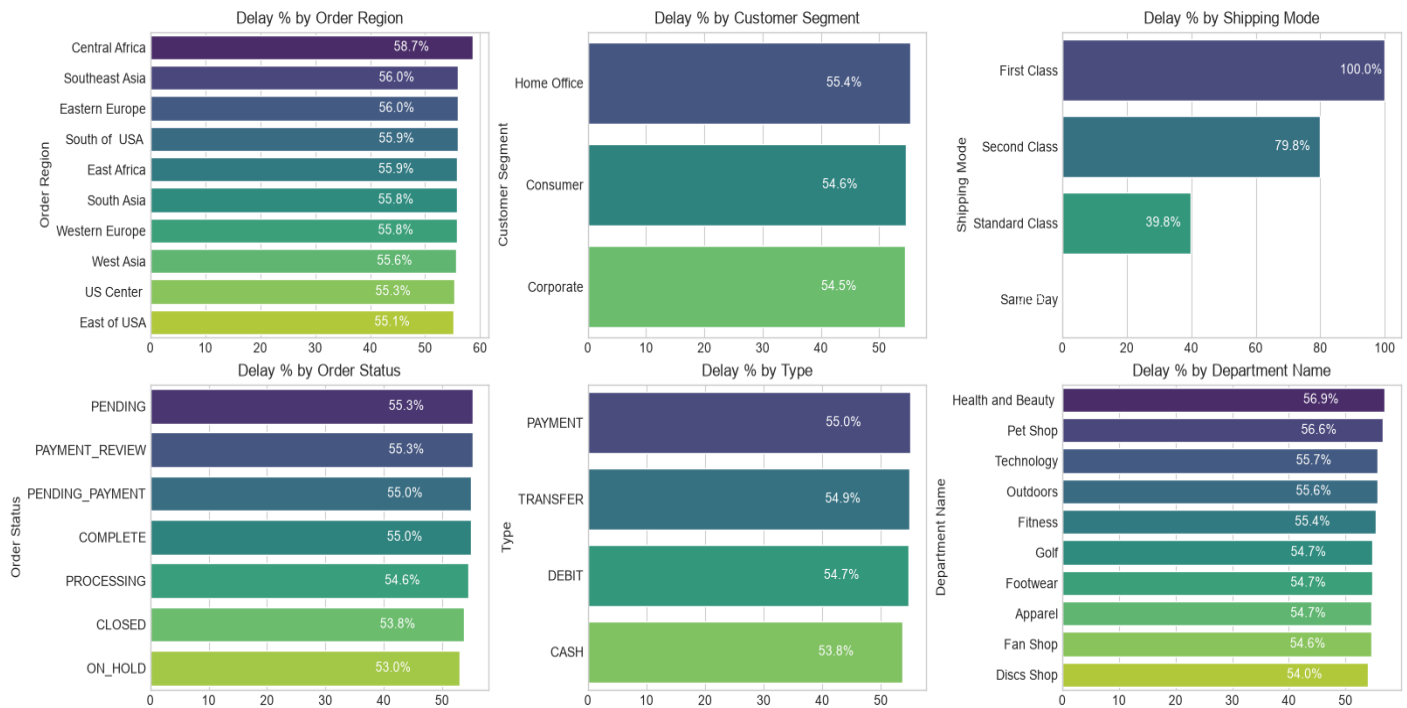
- **Purpose**

The analysis seeks to identify:

- 1) Regions with the highest delay rates.
- 2) Shipping modes with persistent delivery issues.
- 3) Customer segments most impacted by delays.
- 4) Departments associated with poor fulfillment performance.
- 5) Operational processes contributing to delivery failures.

Key Insight

The analysis reveals that delivery performance is not uniformly distributed across the organization. Certain operational segments consistently underperform, indicating targeted opportunities for process improvement.



ROOT CAUSE ANALYSIS

To understand why delays occur, a root cause analysis was performed focusing on regions exhibiting the highest delay percentages.

- 1) Drivers Examined
- 2) Shipping Mode
- 3) Customer Segment
- 4) Department Name
- 5) Order Type
- 6) Order Status
- 7) Findings
- 8) Regional Impact

Regions such as Central Africa and East Africa exhibited significantly higher delay percentages compared to other regions.

- Potential reasons include:
 - 1) Longer shipping distances.
 - 2) Infrastructure limitations.
 - 3) Increased customs and border processing requirements.
 - 4) Limited logistics network coverage.
 - 5) Shipping Mode Influence
 - 6) Standard shipping modes were more susceptible to delays than premium or expedited alternatives.
- Factors include:
 - 1) Longer transit times.
 - 2) Greater dependence on consolidated transportation.
 - 3) Reduced scheduling flexibility.

- 4) Order Status Influence - Orders remaining in pending, processing, or on-hold status demonstrated substantially higher delay rates.

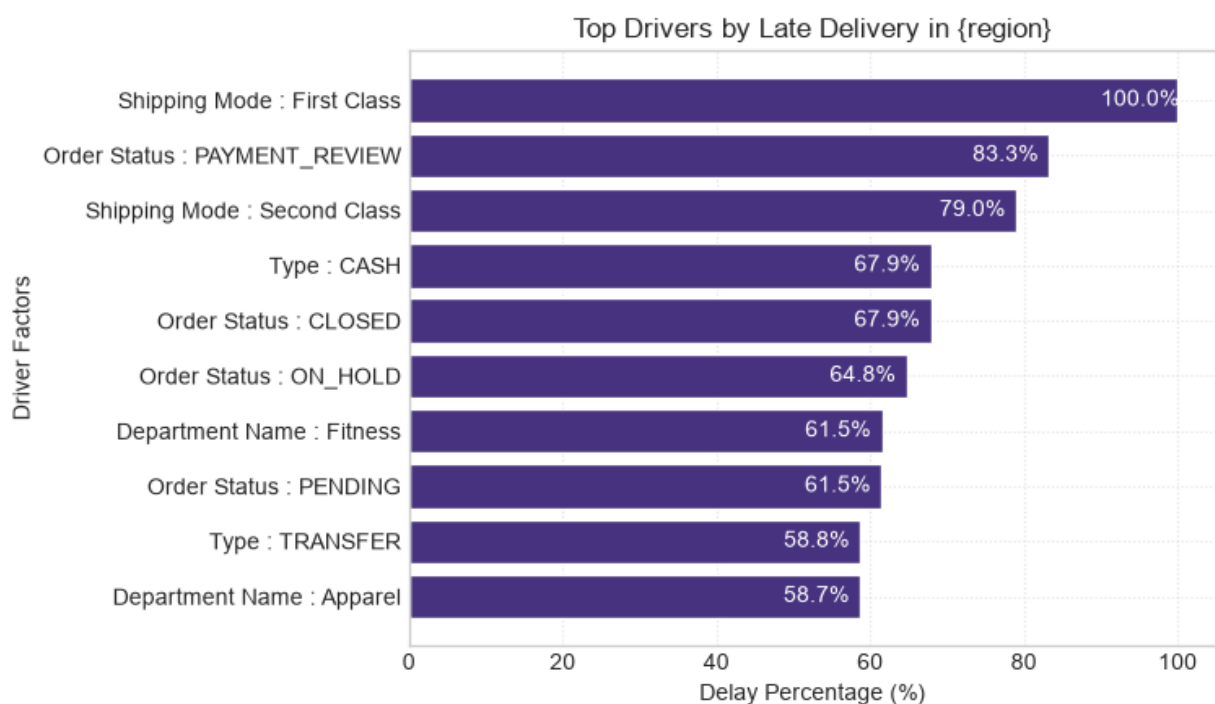
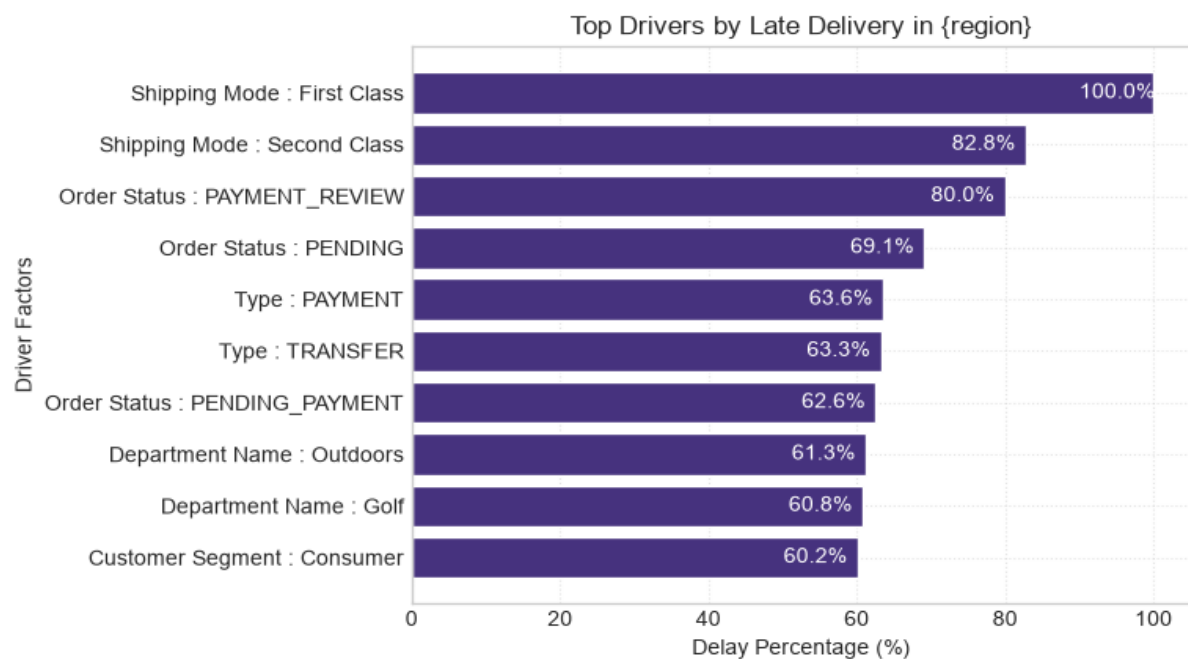
This suggests fulfillment inefficiencies originating before products are physically shipped.

- Department-Level Contribution

Certain product departments showed consistently elevated delay percentages, indicating differences in inventory availability, demand forecasting accuracy, or fulfillment complexity.

Key Insight

Most delays originate from operational process inefficiencies before products enter the carrier network, highlighting the need for improvements in inventory planning, order processing, and warehouse operations.



TIME-BASED DELAY PATTERNS

Time-based analysis was conducted to determine whether delays exhibit seasonal or temporal patterns.

- **Monthly Delay Trends**

Delay percentages were analyzed across all months.

- 1) Findings

Monthly delay rates fluctuate throughout the year, indicating seasonality within fulfillment operations.

- 2) Possible causes include:

- a) Holiday demand surges.
- b) Promotional campaigns.
- c) Inventory replenishment cycles.
- d) Transportation capacity constraints.
- e) Business Implication

Improved forecasting and seasonal capacity planning can reduce delays during high-demand periods.

- **Delay by Day of Week**

Delay percentages were compared across weekdays.

- 1) Findings

Certain weekdays exhibit elevated delay rates, suggesting uneven workload distribution during the week.

- 2) Potential contributing factors include:

- a) Weekday order accumulation.
- b) Staffing limitations.
- c) Carrier scheduling constraints.
- d) Business Implication

Resource allocation should be adjusted to accommodate peak processing periods.

- **Delay by Hour of Day**

Delay rates were evaluated according to order placement hour.

- 1) Findings

Specific order windows demonstrate higher probabilities of late delivery.

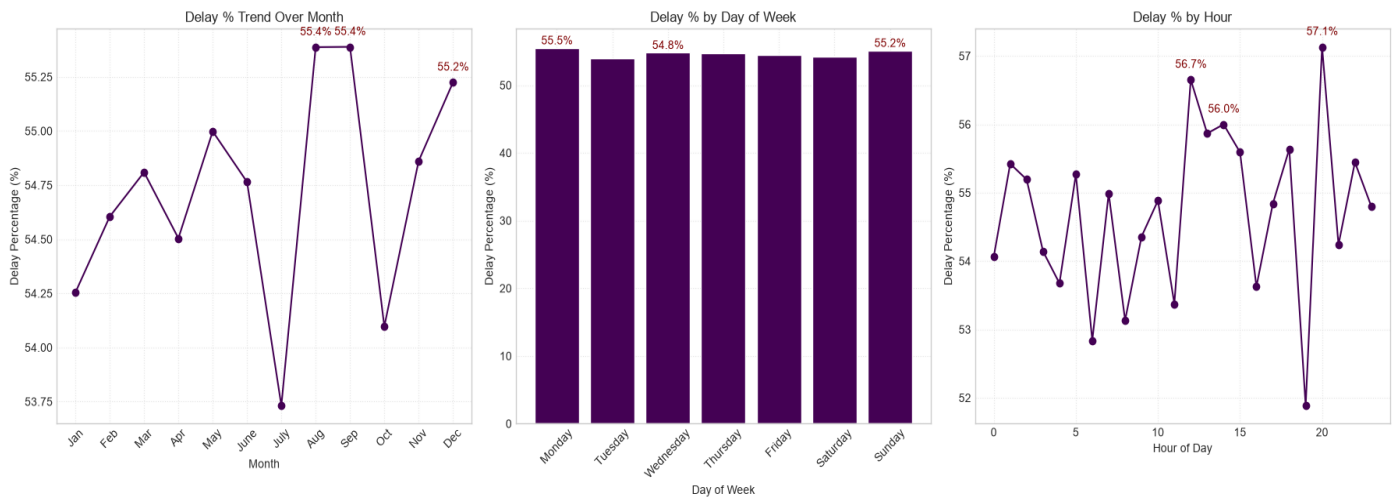
- 2) Reasons may include:

- a) Warehouse cut-off times.
- b) End-of-day processing congestion.
- c) Inventory assignment delays.
- d) Business Implication

Implementing real-time workload balancing and order-prioritization systems can reduce risk during high-volume periods.

Key Insight

Delivery delays display clear temporal patterns, enabling the use of forecasting and scheduling optimization techniques to improve delivery performance.



GEOGRAPHIC DELAY ANALYSIS

- **Objective**

Evaluate delay performance across global regions and countries.

- **Findings**

The analysis identified significant geographic variations in delivery performance. Regions with higher delay percentages exhibit:

- 1) Longer transportation routes.
- 2) Limited logistics infrastructure.
- 3) Greater customs processing requirements.
- 4) Reduced carrier network density.

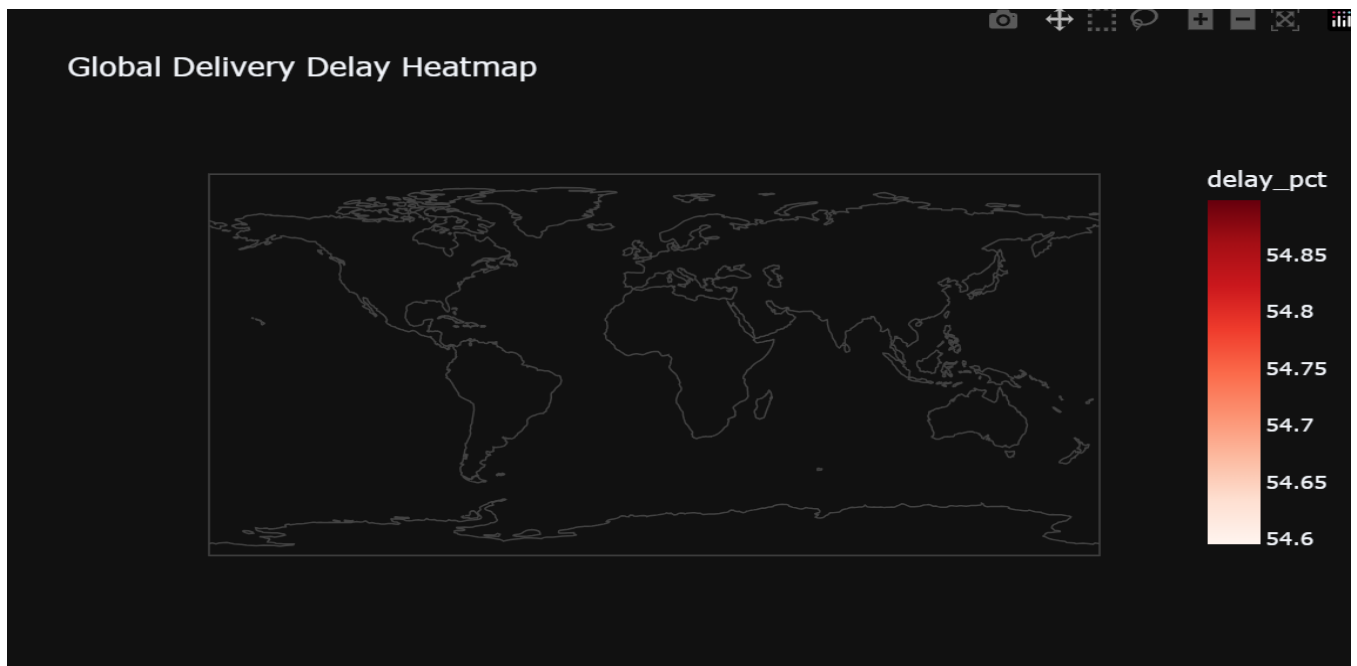
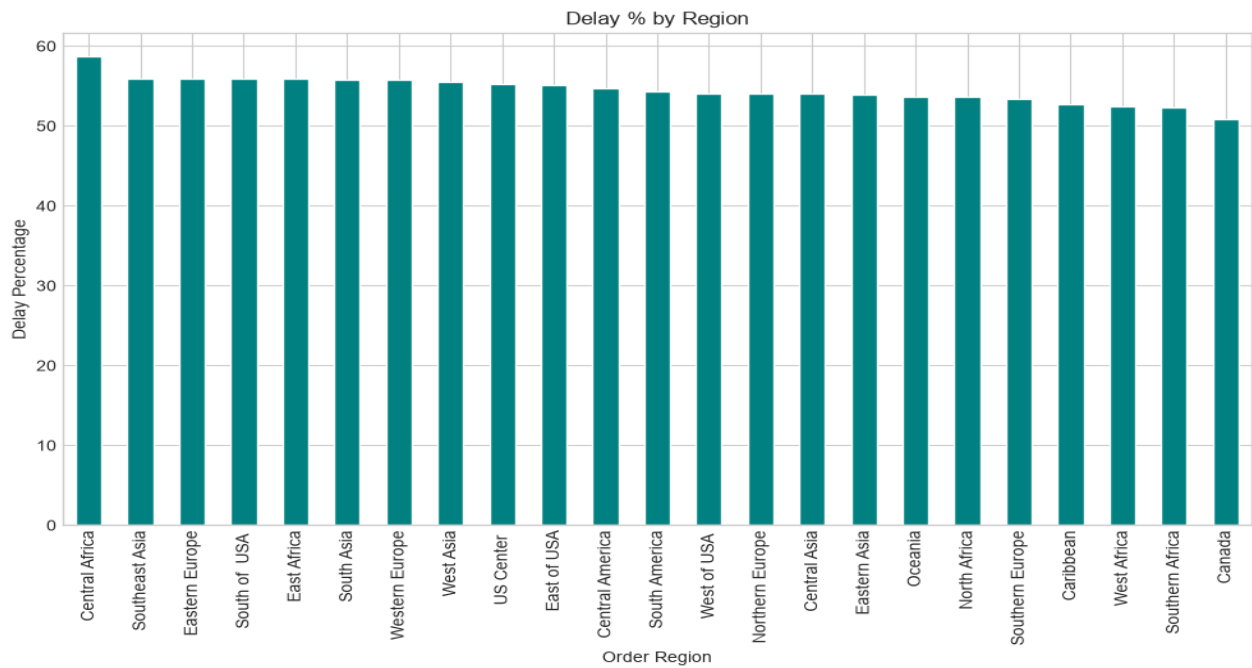
- **Business Implications**

Geographic delay patterns suggest the need for:

- 1) Regional carrier optimization.
- 2) Distributed fulfillment centers.
- 3) Enhanced route planning.
- 4) Country-specific logistics strategies.

Key Insight

Location remains one of the strongest indicators of late-delivery risk and should be incorporated into operational planning and risk-monitoring initiatives.



CUSTOMER SEGMENTATION ANALYSIS

- **Objective**

Identify customer segments exhibiting similar profitability and delivery characteristics.

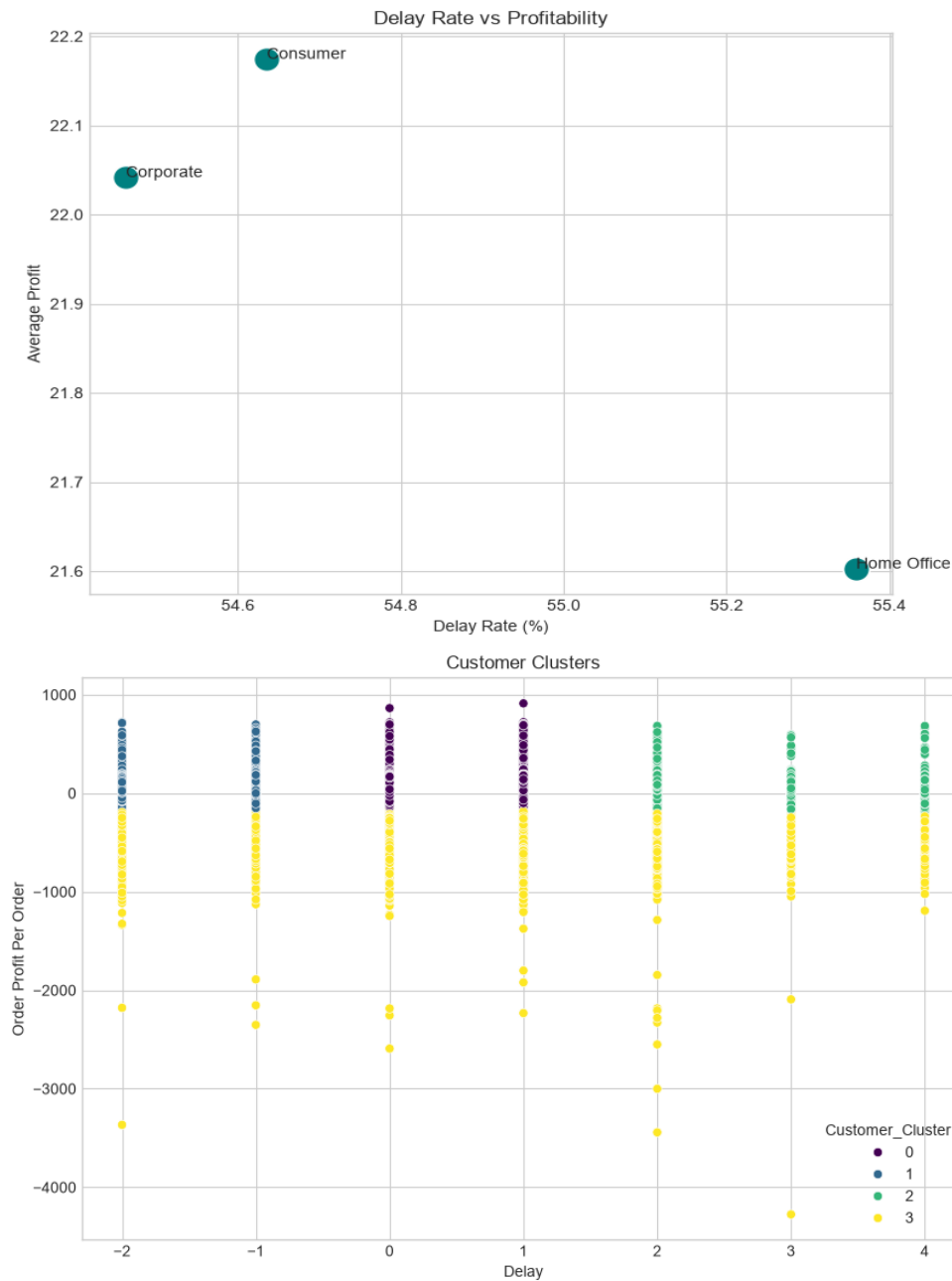
- **Business Interpretation**

The clustering process revealed distinct customer groups:

- 1) Cluster 1 - High profitability and low delays
- 2) Cluster 2 - High profitability but elevated delivery delays
- 3) Cluster 3 - Low profitability and operationally inefficient orders
- 4) Cluster 4 - Moderate profitability with stable delivery performance

Key Insight

Not all delayed customers contribute equally to profitability, allowing operational resources to be allocated more strategically.



MACHINE LEARNING MODEL RESULTS

- **Objective**

Develop a predictive model capable of identifying orders likely to experience late delivery before shipment completion.

- **Target Variable** - Late_delivery_risk

- **Features Used**

- 1) Type
- 2) Days for Shipment (Scheduled)

- 3) Category Name
- 4) Customer Segment
- 5) Department Name
- 6) Order Region
- 7) Shipping Mode
- 8) Order Month
- 9) Order Hour

- **Frequency Encoding**

Categorical variables were encoded using frequency encoding to handle high-cardinality features efficiently.

- **SMOTE Balancing**

Synthetic Minority Oversampling Technique (SMOTE) was applied to address class imbalance within the training dataset.

- **Model Selection**

- 1) A Random Forest Classifier was selected because it:
- 2) Handles complex non-linear relationships.
- 3) Performs well with mixed feature types.
- 4) Is robust to noise and outliers.
- 5) Reduces overfitting through ensemble learning.

- **Model Comparison**

Model	Accuracy
Random Forest	74%
XGBoost	73%

- **Model Performance**

Metric	Result
Model	Random Forest Classifier
Sampling Method	SMOTE
Accuracy	74%

- **Business Applications**

The predictive model can be used to:

- 1) Flag high-risk orders.
- 2) Prioritize fulfillment resources.
- 3) Support proactive customer communication.
- 4) Reduce future late-delivery rates.
- 5) Improve operational planning.

Key Insight

Predictive analytics enables the organization to transition from reactive problem-solving to proactive delivery risk management.

--- XGBoost Classifier ---

Accuracy: 0.73

Precision: 0.89

Recall: 0.59

Classification Report:

	precision	recall	f1-score	support
0	0.62	0.90	0.74	14758
1	0.89	0.59	0.71	19795
accuracy			0.73	34553
macro avg	0.76	0.75	0.72	34553
weighted avg	0.78	0.73	0.72	34553

--- Random Forest Classifier ---

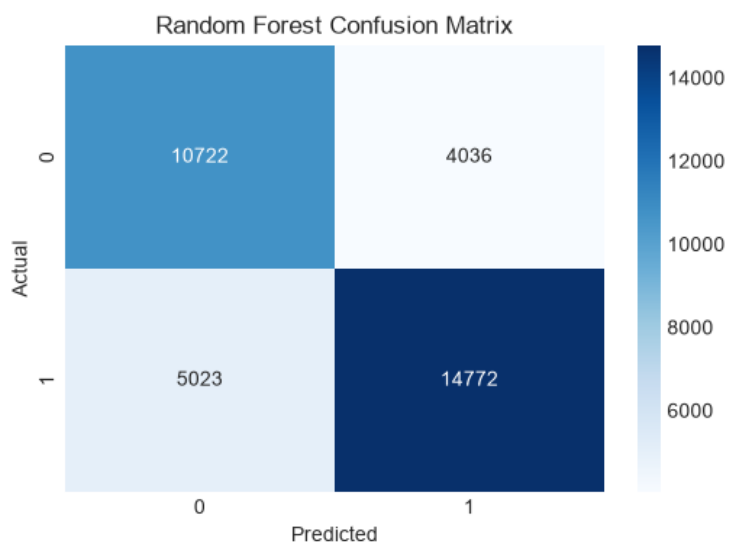
Accuracy: 0.74

Precision: 0.79

Recall: 0.75

Classification Report:

	precision	recall	f1-score	support
0	0.68	0.73	0.70	14758
1	0.79	0.75	0.77	19795
accuracy			0.74	34553
macro avg	0.73	0.74	0.73	34553
weighted avg	0.74	0.74	0.74	34553



MODEL EXPLAINABILITY

- **Objective**

Understand which features contribute most significantly to late-delivery predictions.

- **SHAP Analysis**

SHAP (SHapley Additive exPlanations) was used to interpret model decision-making and quantify feature contributions.

- **Top Drivers**

The most influential predictors include:

- 1) Shipping Mode
- 2) Order Region
- 3) Days for Shipment (Scheduled)
- 4) Customer Segment
- 5) Department Name
- 6) Order Month

- **Business Value**

Explainability provides several advantages:

- 1) Increased stakeholder trust.
- 2) Greater transparency.
- 3) Actionable operational insights.
- 4) More effective risk mitigation.

Key Insight

Late-delivery risk is primarily driven by logistics configuration and regional operating conditions rather than customer characteristics alone.

DELIVERY DELAY FORECASTING

- **Objective**

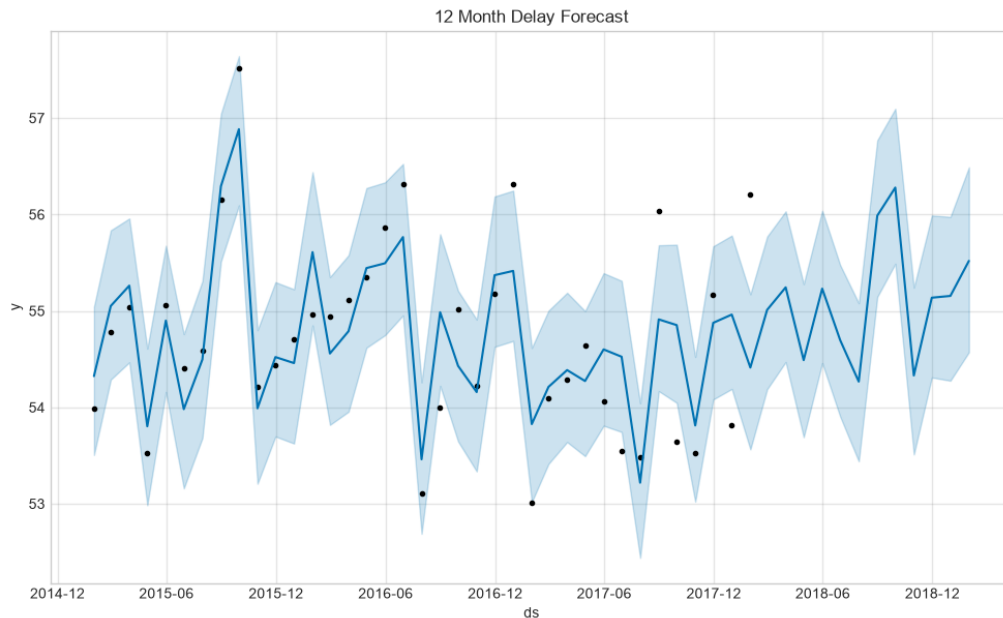
Forecast future delivery-delay trends to support proactive supply-chain planning.

- **Findings**

The forecast indicates that delivery delays are expected to remain persistent if operational processes remain unchanged. Seasonal fluctuations are visible throughout the forecast period, reinforcing previous findings from the time-based analysis.

Key Insight

Operational improvements implemented today can be evaluated against forecasted delay trends to quantify expected performance gains.



SUPPLY CHAIN HEALTH SCORE

A composite Supply Chain Health Score was developed using:

- On-Time Delivery Rate (50%)
- Profitability Rate (30%)
- Model Accuracy (20%)

Supply Chain Health Score: 61.6/100

STRATEGIC RECOMMENDATIONS

1) Focus on High-Risk Regions

Allocate additional resources and logistics support to regions exhibiting the highest delay percentages.

Actions

- Evaluate regional carrier performance.
- Optimize transportation routes.
- Expand local fulfillment capabilities.

2) Improve Shipping Mode Performance

Review shipping options associated with elevated delay rates.

Actions

- Strengthen carrier SLAs.
- Monitor carrier-level KPIs.
- Introduce upgraded shipping alternatives.

3) Optimize Warehouse Operations

Many delays originate during order processing.

Actions

- a) Automate fulfillment workflows.
- b) Improve inventory visibility.
- c) Reduce order processing cycle times.

4) Deploy Predictive Risk Monitoring

Integrate the machine learning model into operational workflows.

Actions

- a) Generate real-time risk scores.
- b) Alert teams to high-risk orders.
- c) Prioritize at-risk shipments.

5) Strengthen Seasonal Planning

Prepare for seasonal spikes proactively.

Actions

- a) Improve demand forecasting.
- b) Increase temporary staffing.
- c) Secure transportation capacity ahead of peak periods.

6) Build Executive Supply Chain Dashboards

This enables continuous monitoring and faster decision-making.

CONCLUSION

This analysis examined 172,765 supply chain orders to assess delivery performance, profitability impact, operational bottlenecks, and predictive delivery risk.

The results indicate that delivery delays are a significant operational challenge, affecting 54.71% of all orders and placing approximately \$2.1 million in profit at risk. Although the organization generated \$7.5 million in profit, recurring fulfillment delays continue to undermine operational efficiency and customer satisfaction.

The investigation identified bottlenecks across regions, shipping modes, departments, and fulfillment processes. Time-based analysis further revealed seasonal and operational patterns that can be leveraged for more effective planning and resource allocation.

Most importantly, the development of a Random Forest model achieving 74% prediction accuracy demonstrates that late-delivery risk can be identified proactively before shipment completion.

By implementing the recommendations outlined in this report, the organization can significantly improve delivery performance, protect profitability, enhance customer experience, and establish a more resilient and data-driven supply chain operation.